Homework of Lie Theory, Spring 2026

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1 Week 1

1. Prove the simplicity of Lie algebras $\mathfrak{sl}_n$, $\mathfrak{o}_n$, $\mathfrak{sp}_{2n}$ and $W(n)$.

2 Week 2

1. Prove that if A is a finite-dimensional associative algebra and for an arbitrary element $a \in A$, $a^n = 0$, then the algebra A is nilpotent. Do not use Wedderburn's theorems. Use Engel's theorem.

2. Let $\operatorname{char} F = 0$; let L be a finite-dimensional solvable Lie algebra; let $e_1, \ldots, e_m$ be a basis of the algebra L. Let M be a finite-dimensional L-module. Suppose that every element e_i, for $1 \leq i \leq m$, acts on M nilpotently. Prove that the algebra L acts on M nilpotently.

3. For every prime p, find an example showing that Lie's theorem fails.

3 Week 4

Exercises are from Humphreys' book.

1. [P21, Ex5] If $x, y \in \operatorname{End} V$ commute, prove that $(x+y)_s = x_s + y_s$, and $(x+y)_n = x_n + y_n$. Show by example that this can fail if x, y fail to commute. [Show first that x, y semisimple (resp. nilpotent) implies $x + y$ semisimple (resp. nilpotent).]

2. [P24, Ex5] Let $L = \mathfrak{sl}(2, \mathbb{F})$. Compute the basis of L dual to the standard basis, relative to the Killing form.

3.[P24, Ex7] Relative to the standard basis of $\mathfrak{sl}(3, \mathbb{F})$, compute the determinant of κ. Which primes divide it?

4.[P30, Ex1] Using the standard basis for $L = \mathfrak{sl}(2, \mathbb{F})$, write down the Casimir element of the adjoint representation of L. Do the same thing for the usual (3-dimensional) representation of $\mathfrak{sl}(3, \mathbb{F})$, first computing dual bases relative to the trace form.

5. [P31, Ex7] It will be seen later on that $\mathfrak{sl}(n, \mathbb{F})$ is actually *simple*. Assuming this and using Exercise 6, prove that the Killing form κ on $\mathfrak{sl}(n, \mathbb{F})$ is related to the ordinary trace form by $\kappa(x, y) = 2n\operatorname{Tr}(xy)$.

4 Week 7

Exercises are from Humphreys' book.

1. [P34, Ex2] $M = \mathfrak{sl}(3, \mathsf{F})$ contains a copy of L in its upper left-hand 2×2 position. Write M as direct sum of irreducible L-submodules (M viewed as L-module via the adjoint representation): $V(0) \oplus V(1) \oplus V(1) \oplus V(2)$.

2. [P34, Ex6] Decompose the tensor product of the two L-modules $V(3)$, $V(7)$ into the sum of irreducible submodules: $V(4) \oplus V(6) \oplus V(8) \oplus V(10)$. Try to develop a general formula for the decomposition of $V(m) \otimes V(n)$.

3. [P41, Ex1] If L is a classical linear Lie algebra of type A_ℓ, B_ℓ, C_ℓ, or D_ℓ (see (1.2)), prove that the set of all diagonal matrices in L is a maximal toral subalgebra, of dimension ℓ. (Cf. Exercise 2.8.)

4. [P41, Ex10] Prove that no four, five or seven dimensional semisimple Lie algebras exist.

5 Week 10

1. [P46, Ex2] Prove that $\Phi^\vee$ is a root system in E, whose Weyl group is naturally isomorphic to $\mathcal{W}$; show also that $\langle \alpha^\vee, \beta^\vee \rangle = \langle \beta, \alpha \rangle$, and draw a picture of $\Phi^\vee$ in the cases $\mathsf{A}_1, \mathsf{A}_2, \mathsf{B}_2, \mathsf{G}_2$.

2. [P54, Ex1] Let $\Phi^\vee$ be the dual system of $\Phi, \Delta^\vee = \{\alpha^\vee | \alpha \in \Delta\}$. Prove that $\Delta^\vee$ is a base of $\Phi^\vee$. [Compare Weyl chambers of Φ and $\Phi^\vee$.]

3. [P54, Ex10] Given $\Delta = \{\alpha_1, \dots, \alpha_l\}$ in Φ, let $\lambda = \sum_{i=1}^{l} k_i \alpha_i$ ($k_i \in \mathbf{Z}$, all $k_i \geq 0$ or all $k_i \leq 0$). Prove that either λ is a multiple (possibly 0) of a root, or else there exists $\sigma \in \mathcal{W}$ such that $\sigma\lambda = \sum_{i=1}^{l} k_i' \alpha_i$, with some $k_i' > 0$ and some $k_i' < 0$. [Sketch of proof: If λ is not a multiple of any root, then the hyperplane P_λ orthogonal to λ is not included in $\bigcup_{\alpha \in \Phi} P_\alpha$. Take $\mu \in P_\lambda - \bigcup_{\alpha \in \Phi} P_\alpha$. Then find $\sigma \in \mathcal{W}$ for which all $(\alpha_i, \sigma\mu) > 0$. It follows that $0 = (\lambda, \mu) = (\sigma\lambda, \sigma\mu) = \sum k_i'(\alpha_i, \sigma\mu)$.]

4. Prove that if α, β are roots, $\beta \neq \pm\alpha$, k is a real number and $\beta + k\alpha$ is a root, then k is an integer.

6 Week 11

1. Find Weyl groups of $\mathsf{A}_2, \mathsf{B}_2, \mathsf{G}_2$.

Let $\Delta \subset \mathbb{R}^n$ be a root system and let $B = \{\beta_1, \dots, \beta_n\}$ be a base of Δ. The Cartan matrix is defined as

$$\left(\frac{2(\beta_i | \beta_j)}{(\beta_j | \beta_j)} \right)_{1 \leq i,j \leq n}.$$

2. Prove that the Cartan matrix of a root system is positive definite.
3. Find all Cartan matrices of rank 3.
4. Show that a Cartan matrix determines the root system uniquely up to isomorphism.